

CHRONIC SUPPURATIVE MASTITIS MIMICKING BREAST CANCER IN A NON PUERPERAL WOMAN: A CASE REPORT

S. Hayuningsih^{1,4}, T. W. Handarini², L. Choridah³

¹Fellowship Women Imaging Academic Hospital Gadjah Mada University
²Soetomo Academic General Hospital
³Faculty of Medicine, Public Health, and Nursing Gadjah Mada University/ Dr. Sardjito General Hospital
⁴Professor Dr. W. Z. Johannes Regional General Hospital, Kupang, East Nusa Tenggara, Indonesia



BACKGROUND

Chronic suppurative mastitis (CSM) is a rare benign inflammatory breast disease characterized by persistent infection, chronic inflammation, and abscess formation. It is clinical and radiological features may closely mimic breast carcinoma, making histopathological examination essential for definitive diagnosis.

OBJECTIVES

Primary: To illustrate the radiologic–pathologic discrepancy in a non-puerperal CSM case and emphasise the role of histopathology for definitive diagnosis.

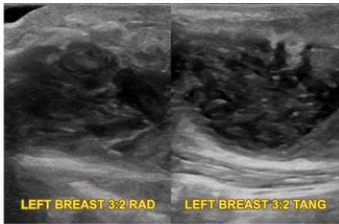
Secondary: To summarise imaging patterns that may help distinguish CSM from breast carcinoma in everyday practice.

METHODS

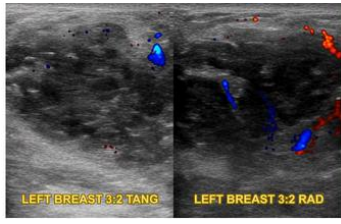
Case Report:
A 23 year old non puerperal woman with no history of breastfeeding presented with a one-month history of painful swelling and mild erythema of the left breast, referred from Oncologist Surgeon, as suspected Malignancy.

RESULTS

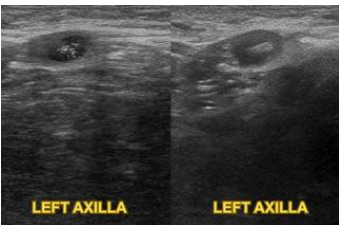
Breast ultrasonography demonstrated an irregular hypoechoic mass, wider than taller, with echogenic material, without internal moving echo, accompanied by soft tissue edema and left axillary lymphadenopathy, raising concern for breast malignancy. Chest radiography revealed a left breast soft tissue mass without pulmonary abnormalities. However, histopathological examination confirmed chronic suppurative mastitis without evidence of tuberculosis and malignancy.



An irregular hypoechoic mass is observed in the left breast at the 3 o'clock position, 2 cm from the nipple, wider than taller, and microlobulated margins, with dimensions of 3.99 x 2.85 x 3.50 cm



The mass has echogenic material, without internal moving echo, accompanied by soft tissue edema, and vascularization within and at the periphery of the mass.



Left axillary lymphadenopathy is observed.



Chest radiograph reveals a soft tissue mass in the left breast, the heart and lungs are normal, and there is mild thoracic dextroscoliosis.

Key numerical findings

A suspicious left breast mass at the 3 o'clock position, 2 cm from the nipple, wider than taller, and microlobulated margins, with dimensions of 3.99 x 2.85 x 3.50 cm, accompanied by soft tissue edema and left axillary lymphadenopathy.

CONCLUSION / TAKE-HOME MESSAGE

Chronic suppurative mastitis should be considered in the differential diagnosis of suspicious breast masses, even in young non puerperal women without a history of breastfeeding.

Histopathological confirmation is essential for establishing the correct diagnosis, avoiding unnecessary oncologic intervention, and guiding appropriate management.

Accurate diagnosis requires close multidisciplinary collaboration among radiologists, surgeons, and pathologists.

REFERENCES

- Mary S Guirguis, Beatriz Adrada, Lumarie Santiago, Rosalind Candelaria, Elsa Aribas. Mimickers of breast malignancy: imaging findings, pathologic concordance and clinical management. *Insights Imaging*. 2021;12:40.
- Rella R, et al. A pictorial review on mastitis: Clinical aspects, imaging features and complications. *Journal of Imaging*. 2026;12(5):181.
- Tariq Rashid, MD , Tiffany M. Sae-Kho, MD , Kara L. Heuvelhorst, DO , Katrina N. Glazebrook, MB, ChB. Breast imaging of infectious disease. *Br J Radiol*. 2023;96:20220649.
- American College of Radiology. *ACR BI-RADS® Atlas: Breast Imaging Reporting and Data System*. 5th ed. Reston, VA: American College of Radiology; 2013.
- Robert T Fazzio, Sejal S Shah, Nicole P Sandhu, Katrina N Glazebrook. Idiopathic granulomatous mastitis: imaging update and review. *Insights Imaging*. 2016;7:531–539.

ACKNOWLEDGEMENTS

Funding: None
Collaborators: Radiology, Surgery, Pathology teams
Conflict of Interest: None

